MENTAL HEALTH & LIFESTYLE HABITS

Data Analytics Project Report

Business Understanding • Data Engineering (ETL) • Power BI Dashboard

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1. Introduction

This report documents the complete workflow carried out for the Mental Health & Lifestyle Habits data analytics project, from initial business understanding of the dataset, through the data engineering (ETL) pipeline that structured the data into a data warehouse, to the final interactive Power BI dashboards built for analysis and reporting.

The project follows a standard analytics pipeline: raw data was first explored and understood, then transformed and loaded into a dimensional (star schema) model using SQL Server Integration Services (SSIS), and finally visualized through a multi-page Power BI dashboard covering demographics, mental health indicators, and lifestyle habits.

1.1 Project Scope

- Business Understanding: exploring and statistically profiling the raw dataset.
- Data Engineering: building an ETL pipeline and a star-schema data warehouse.
- Dashboarding: designing an interactive multi-page Power BI report.

2. Business Understanding

The Mental Health & Lifestyle Habits dataset is a structured dataset built for exploring the relationship between everyday lifestyle behaviors and self-reported mental health status. It contains 50,000 individual respondent records, capturing demographic information alongside mental health indicators and lifestyle habits such as sleep, work hours, physical activity, social media use, diet, smoking, alcohol consumption, and medication use.

The dataset is well suited for exploratory data analysis, demographic profiling, and dashboard/report creation, and is fully compatible with BI tools such as Power BI. Each column is clean, consistently formatted, and free of missing demographic values.

2.1 Dataset at a Glance

Metric	Value
Total Records	50,000
Total Variables	17
Missing Demographic Values	None
Duplicate Records	None
Age Range	18 – 65 years
Respondents Reporting a Mental Health Condition	24,997 (50.0%)
Respondents With No Reported Condition	25,003 (50.0%)

2.2 Columns in the Dataset

#	Column	Description
1	User_ID	Unique identifier for each respondent (1 to 50,000)
2	Age	Respondent age, 18–65 years
3	Gender	Male, Female, Non-binary, or Prefer not to say
4	Occupation	IT, Healthcare, Education, Finance, Engineering, Sales, Other
5	Country	USA, UK, Canada, Australia, Germany, India, or Other
6	Mental_Health_Condition	Whether the respondent reports a mental health condition (Yes/No)
7	Severity	Low, Medium, or High — recorded only when a condition is present
8	Consultation_History	History of consulting a mental health professional (Yes/No)
9	Stress_Level	Self-reported stress level: Low, Medium, or High
10	Sleep_Hours	Average hours of sleep per night (4–10 hours)
11	Work_Hours	Average work hours per week (30–80 hours)
12	Physical_Activity_Hours	Hours of physical activity per week (0–10 hours)
13	Social_Media_Usage	Average hours of social media use per day (0.5–6 hours)
14	Diet_Quality	Healthy, Average, or Unhealthy

#	Column	Description
15	Smoking_Habit	Non-Smoker, Occasional, Regular, or Heavy Smoker
16	Alcohol_Consumption	Non-Drinker, Social, Regular, or Heavy Drinker
17	Medication_Usage	Currently taking medication related to mental health (Yes/No)

2.3 Demographic Profile

The respondent base is broad and evenly spread across age, gender, occupation, and country, with no single dominant subgroup — making the dataset well balanced for cross-category comparison.

- Age: ranges from 18 to 65 years, mean 41.5, median 41, approximately uniform distribution.
- Gender: split almost evenly across Female (25.3%), Prefer not to say (25.1%), Male (24.9%), Non-binary (24.7%).
- Occupation: 7 categories, each roughly one-seventh of respondents (IT, Healthcare, Education, Finance, Engineering, Sales, Other).
- Country: 7 country groupings, nearly evenly distributed (USA, UK, Canada, Australia, Germany, India, Other).

2.4 Mental Health Indicators

- Mental Health Condition: exactly half of respondents (24,997 / 50,000 = 50.0%) report a condition.
- Severity: recorded only for the 24,997 condition cases, split almost equally between Low, Medium, and High.
- Stress Level: split almost evenly across Low, Medium, and High.
- Consultation History & Medication Usage: ~49.8% report consultation history and ~49.7% report current medication use, both close to a 50/50 split regardless of condition status.

2.5 Lifestyle Habits

- Sleep, Work Hours, and Physical Activity are all close to uniformly distributed (Sleep: 4–10h, mean 7.0h; Work: 30–80h, mean 55.1h; Activity: 0–10h, mean 5.0h).
- Social Media Usage ranges from 0.5 to 6 hours/day, mean 3.2 hours.
- Diet Quality, Smoking Habit, and Alcohol Consumption are each split almost evenly across their categories (33.3–33.5%, 24.9–25.2%, and 25.0–25.1% respectively).

2.6 Key Relationships Between Variables

Every lifestyle and demographic variable was cross-tabulated against Mental_Health_Condition, using chi-square tests for categorical pairs and Pearson correlation for numeric pairs. The finding: across every variable tested, there is no statistically meaningful relationship with mental health condition, severity, or stress level — the condition rate stays at essentially 50% no matter how the data is sliced.

Variable Pair	p-value	Result
Gender × Condition	0.352	Not significant

Variable Pair	p-value	Result
Occupation × Condition	0.151	Not significant
Country × Condition	0.965	Not significant
Diet Quality × Condition	0.225	Not significant
Smoking Habit × Condition	0.232	Not significant
Alcohol Consumption × Condition	0.556	Not significant
Stress Level × Condition	0.940	Not significant
Consultation History × Condition	0.914	Not significant

Pearson correlation coefficients among the numeric variables (Age, Sleep_Hours, Work_Hours, Physical_Activity_Hours, Social_Media_Usage) are all effectively zero (between -0.01 and 0.01), confirming these variables were generated independently of one another.

2.7 What This Means for Analysis

- The dataset behaves as a randomized, uniformly distributed sample rather than one built around designed cause-and-effect relationships.
- It is well suited for descriptive statistics, data cleaning, segmentation, and visualization practice.
- It is not suited for predictive modeling or causal analysis of mental health outcomes, since no variable meaningfully predicts another.

2.8 Use Cases

- Exploratory Data Analysis (EDA): summarizing, grouping, and visualizing demographic and lifestyle data at scale.
- Dashboard & Report Creation: building interactive dashboards covering demographics, mental health indicators, and lifestyle habits.
- Data Cleaning Practice: handling the Severity column's conditional missingness.
- Statistical Testing Practice: chi-square tests, correlation analysis, and hypothesis testing on a dataset with a known null result.
- Teaching & Training: demonstrating the difference between a dataset with designed relationships and one without.

2.9 Data Quality Notes

- No missing values in any demographic or lifestyle column; no duplicate User_ID values across all 50,000 records.
- The Severity column is conditionally populated: blank for the 25,002 respondents without a clearly recorded condition, populated with Low/Medium/High for the remaining 24,997 — expected structural missingness, not a data quality defect.
- All numeric variables fall within sensible real-world bounds with no outliers or negative values.
- Category distributions across every categorical column are close to uniform, consistent with a synthetically generated or randomly sampled dataset.

3. Data Engineering

To prepare the raw dataset for analysis and reporting, an ETL (Extract, Transform, Load) pipeline was built using SQL Server Integration Services (SSIS), loading the cleaned and structured data into a dimensional (star-schema) data warehouse.

3.1 Data Warehouse Schema (Star Schema)

The data model follows a star-schema design, centered on a single fact table connected to three dimension tables, plus a dedicated measures table used for DAX calculations in Power BI:

Table	Type	Purpose
Fact_MentalHealth	Fact table	Central table holding the measurable/quantitative respondent-level data (stress level, sleep hours, work hours, severity, medication usage, etc.) linked to all dimensions.
Dim_Demographics	Dimension	Respondent demographic attributes: Age, Gender, Occupation, Country.
Dim_HealthStatus	Dimension	Mental health status attributes: Mental_Health_Condition, Severity, Consultation_History — modeled as a Slowly Changing Dimension (SCD).
Dim_Lifestyle	Dimension	Lifestyle attributes: Sleep_Hours, Work_Hours, Physical_Activity_Hours, Social_Media_Usage, Diet_Quality, Smoking_Habit, Alcohol_Consumption.
_Measures	Measures table	A dedicated (empty) table holding all DAX measures used across the dashboard, keeping calculations organized and separate from the data tables.

3.2 ETL Pipeline — SSIS Data Flow

The health status dimension (DimHealthStatus) is built and maintained using an SSIS Data Flow Task designed as a Slowly Changing Dimension (SCD), so historical changes to a respondent's health status are tracked rather than overwritten. The pipeline stages are as follows:

- Flat File Source: reads the raw source data file (the respondent-level CSV extract).
- Data Conversion: converts source columns to the correct data types required by the destination warehouse table.
- Derived Column: computes/adds calculated columns needed before the dimension load.
- Slowly Changing Dimension (SCD): compares incoming records against the existing dimension table and routes rows into two paths — Historical Attribute Inserts (for changed records) and New Output (for brand-new records).
- Derived Column 1: applies transformation logic to the historical/changed records path before update.
- OLE DB Command: executes the update (UPDATE statement) against existing dimension rows for changed records.
- Union All: merges the updated historical records with the newly inserted records into a single stream.

- Derived Column 2: final column-level adjustments applied to the unified stream.
- Insert Destination: loads the final, transformed records into the Dim_HealthStatus table in the data warehouse.

3.3 Why a Slowly Changing Dimension?

Mental health status (condition, severity, consultation history) can change for a respondent over time. Using an SCD pattern ensures that when a respondent's health status changes, the previous state is preserved as history (via the Historical Attribute Inserts path) rather than being lost, while genuinely new respondents are simply inserted (via the New Output path). This keeps Dim_HealthStatus both accurate and historically traceable for trend analysis in the dashboard.

4. Dashboard & Visualization (Power BI)

The final data warehouse feeds a multi-page interactive Power BI report, built on top of the star-schema model described above. The report is organized into four main analytical pages plus supporting exploration pages, using slicers, KPI cards, and a range of chart types to let users explore the data from different angles.

4.1 Overview Page

The landing page presents top-level KPIs for the full respondent base — average age, total patients, heavy smoker percentage, insomnia rate, and average work hours — alongside a breakdown of patients by country and stress level, an age-group distribution, and a world map showing total patients by country.

4.2 LifeStyle Page

This page focuses on daily lifestyle habits: average sleep hours, burnout risk, average social media usage, and high-stress percentage as KPIs, plus alcohol consumption among heavy smokers, patients by diet quality and stress level, sleep category distribution, social media usage by age group and country, and a smoking-habit-by-age pivot table. Slicers allow filtering by country, age, and severity level.

4.3 According to Lifestyle Page

This page cross-references lifestyle patterns with mental health outcomes: average sleep and work hours by gender, mental health condition share by occupation, average sleeping hours by age group, and stress level by age group.

4.4 Health Status Page

This page focuses on clinical/consultation indicators: distribution of medication usage by stress level, severity distribution across consultations, total patients and medication usage percentage by severity and smoking habit, and a summary of user counts, condition rate, consultation count, and medication usage count. A severity slicer allows filtering the whole page.

5. Conclusion

This project walked the full analytics pipeline for the Mental Health & Lifestyle Habits dataset: understanding and statistically profiling the business/data context, engineering a star-schema data warehouse with a properly maintained Slowly Changing Dimension for health status, and delivering a multi-page interactive Power BI dashboard covering demographics, lifestyle habits, and mental health indicators.

The statistical analysis confirms that the dataset is uniformly distributed with no significant relationships between lifestyle/demographic variables and mental health outcomes — making it best suited for descriptive analytics, dashboarding, and data engineering practice rather than causal or predictive modeling. The resulting dashboard nonetheless demonstrates a complete, professional BI workflow from raw data to decision-ready visuals.

6. Team

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DAX Measures

The Power BI solution uses a dedicated _Measures table to organize DAX calculations.

Measure	Purpose
Total Patients	Counts respondents
Average Age	Average age
Average Sleep Hours	Average sleep
High Stress %	High stress percentage
Medication Usage %	Medication percentage
Condition Rate %	Condition percentage